ASSIGNMENT – 1

Task – 1: Linux Flavours Research

SOLUTION-

There are various Linux flavours version are created each have specific and different its own features, target audience, and package management system.

Some flavours are enlisted below-

1. Ubuntu

- **Target Audience:** Beginners, students, and developers.
- **Key Features:**
  - ➤Easy to install and use.
  - ➤Large community support.
  - ➤Regular updates and security patches.
- **Package Manager:** APT (apt).

2. CentOS

- **Target Audience:** System administrators and server users.
- **Key Features:**
  - ➤Stable and reliable operating system.
  - ➤Suitable for web and enterprise servers.
  - ➤Long-term support.
- **Package Manager:** YUM / DNF.

3. Red Hat Enterprise Linux

- **Target Audience:** Enterprises and large organizations.
- **Key Features:**
  - ➤Commercial support from Red Hat.
  - ➤High security and stability.
  - ➤Widely used in enterprise environments.

- **Package Manager:** DNF / YUM.

4. Debian

- **Target Audience:** Developers and advanced users.
- **Key Features:**
  - ➤Highly stable and secure.
  - ➤Large software repository.
  - ➤Strong open-source community support.
- **Package Manager:** APT (apt).

5. Fedora

- **Target Audience:** Developers and Linux enthusiasts.
- **Key Features:**
  - ➤Includes the latest software technologies.
  - ➤Strong security features.
  - ➤Sponsored by Red Hat.
- **Package Manager:** DNF.

6. Arch Linux

- **Target Audience:** Advanced and experienced Linux users.
- **Key Features:**
  - ➤Highly customizable.
  - ➤Lightweight and fast.
  - ➤Rolling release model with continuous updates.
- **Package Manager:** Pacman.

Task – 2: What is Linux?

SOLUTION-

Linux is a free and open-source operating system that is widely used on computers, servers, smartphones, and cloud platforms. It is known for its stability, security, and flexibility. Today, Linux is one of the most popular operating systems in the world and is commonly used in the fields of software development, networking, cloud computing, and DevOps. The history of Linux began in 1991. Linus Torvalds created a new operating system kernel as a personal project. He wanted an operating system that was free and could be improved by anyone. Linus released the source code to the public, allowing developers worldwide to contribute to its development.

KERNEL-

The Linux kernel is the core part of the operating system. It acts as a bridge between computer hardware and software applications. The kernel manages system resources such as memory, processors, storage devices, and input/output operations. Without the kernel, software cannot communicate with the hardware.

Linux is a powerful, reliable, and open-source operating system built around the Linux kernel. Its combination with GNU tools created a complete operating system that is now widely used across various industries. Due to its flexibility, security, and community-driven development, Linux remains one of the most important operating systems in modern computing.

Task – 3: Linux vs Windows vs macOS Comparison

SOLUTION-

Linux

- Linux is a free and open-source operating system.
- It is highly secure and less affected by viruses.
- Users can customize almost every part of the system.
- It performs well even on older hardware.
- Linux is widely used in servers, cloud computing, and DevOps.

- It has a large collection of free software.
- Gaming support is improving but is not as extensive as Windows.

Windows

- Windows is a commercial operating system developed by Microsoft.
- It is the most widely used operating system for personal computers.
- It provides excellent support for software, games, and hardware devices.
- The interface is user-friendly and easy to learn.
- Windows is more vulnerable to malware and viruses than Linux.
- Most businesses and home users prefer Windows.
- Regular updates provide new features and security improvements.

MACOS

- macOS is an operating system developed by Apple for Mac computers.
- It offers a smooth and user-friendly experience.
- It is known for its strong security and stability.
- macOS is optimized to work efficiently with Apple hardware.
- It is popular among designers, video editors, and creative professionals.
- It supports many productivity and multimedia applications.
- Hardware support is limited to Apple devices.

Task – 4: Why Linux?

SOLUTION-

Because Linux is preferred in many real-world environments because it is secure, stable, flexible, and cost-effective. Its ability to be customized and optimized for different tasks makes it the first choice for servers, cloud computing, embedded systems, supercomputers, and DevOps operations.

USE CASE ANALYSIS-

1. Web Servers

Linux is the most popular operating system for web servers. It is stable, secure, and can run continuously for long periods without restarting. Most web server software such as Apache and Nginx works efficiently on Linux.

2. Cloud Computing

Linux is widely used in cloud platforms because it is lightweight and cost-effective. Major cloud providers such as AWS, Microsoft Azure, and Google Cloud offer Linux-based virtual machines. Its flexibility makes it ideal for managing cloud resources.

3. Embedded Systems

Linux is commonly used in embedded devices such as smart TVs, routers, ATMs, and IoT devices. It can be customized to meet specific hardware requirements. Its low resource usage makes it suitable for small devices.

4. Supercomputers

Most of the world's supercomputers run Linux because of its high performance and reliability. Linux allows researchers and scientists to customize the system according to their computational needs. It efficiently handles large-scale processing tasks.

5. DevOps Environments

Linux is the preferred operating system for DevOps because most DevOps tools are designed to work on Linux. Tools such as Docker, Kubernetes, Jenkins, and Ansible run smoothly on Linux systems. It provides powerful command-line utilities and automation capabilities.

Task – 5: Linux Components Diagram

SOLUTION-

Linux Components Diagram

Linux Components Diagram

```
+---------------------+
|  User Applications  |
| (Browser, Editor,   |
|  Media Player, etc.) |
+----------+----------+
           |
           v
+---------------------+
|       Shell         |
| (Bash, Zsh, etc.)   |
+----------+----------+
           |
           v
+---------------------+
|      Kernel         |
| (Core of Linux OS)  |
+----------+----------+
           |
           v
+---------------------+
|   Hardware Layer    |
| CPU, RAM, Disk,     |
| Keyboard, Network   |
+---------------------+
```

1.Hardware Layer

The hardware layer consists of physical components such as the CPU, memory, storage devices, keyboard, and network devices. These components provide the resources required for the system to operate.

2. Kernel

The kernel is the core component of Linux. It manages hardware resources, memory, processes, and device communication. It acts as a bridge between software and hardware.

3. Shell

The shell is a command interpreter that allows users to interact with the operating system. It receives commands from users and passes them to the kernel for execution.

4. User Applications

User applications are programs such as web browsers, text editors, media players, and office software. These applications use the shell and kernel services to access hardware resources.